

NCHO Patient-Link: A Patient-Managed Digital
Health Record System
Frontend Web Development Plan
Patient-Link

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"Linking patients to better care."

The purpose of the NCHO Patient-Link application is to streamline the Out-Patient Department (OPD) flow at the Naga City Health Office through a secure digital health record system.

Patients carry a digital Health Passport via QR code, removing dependency on physical folders and repetitive paper forms. Access is controlled through role-based authentication, provider 2FA, and system-issued session controls. The system is built to enforce Data Privacy Act compliance while giving providers fast access to baseline patient data for triage and consultation.

Goal

The primary objective of the NCHO Patient-Link project is to bridge patient and provider information flow through a secure and operationally efficient frontend experience. The system provides a unified login path, a read-only patient mobile dashboard, and a provider portal focused on rapid clinical data entry. Access to patient records is controlled by QR-based identity matching and authenticated provider sessions. Automated audit logging is embedded in provider-facing API flow and exposed to patients through the Privacy and Access Log.

Objectives

- Provide NCHO staff with rapid retrieval of patient information through QR scan and patient lookup flows that open the provider clinical record view without manual chart retrieval.
- Provide patients with a read-only Health Passport interface where they can view profile data, encounter history, and uploaded medical records without any patient-side edit capability.
- Enforce Data Privacy Act-aligned accountability by automatically recording provider access and clinical actions into audit logs that feed patient-facing privacy visibility and admin monitoring.

Scope

Inclusions (What the project will deliver)

- Unified role-based login for Patients and Providers with protected routes.
- Patient Health Passport (read-only): QR display, baseline profile, recent visits, records, and privacy logs.
- Provider Patient Directory with QR scan and manual lookup to open patient clinical record view.
- Encounter workflow: start/resume session, encode vitals and clinical notes, finalize visit, and store encounter history.
- Medical Records Archive upload workflow for authorized providers (PDF/image clinical files).

- Automated audit logging on protected provider workflow endpoints with patient/provider attribution.
- Patient-facing Privacy and Access Log and admin-facing audit feed sourced from AUDIT_LOGS.

Exclusions (What is not included in this phase)

- No patient-side create, update, or delete capability for clinical data.
- No anonymous/public access to protected health records.
- No external EMR/HIS/LIS integration or third-party health system synchronization.
- No telemedicine/video consultation module.
- No billing, claims, or insurance reimbursement processing.
- No mobile-native app release (current implementation is web frontend + API backend).
- No AI diagnostic recommendation engine.

Project overview

NCHO Patient-Link is a healthcare management platform designed to support patient registration, clinical encounter documentation, medical record archiving, and privacy tracking. The system enforces patient-facing read-only access, provider-authenticated write access, and middleware-level activity logging. Clinical encounter completion, document upload, and patient record access are handled through authenticated backend transactions.

Target Audience

The system serves authorized healthcare providers (Doctors, Nurses, Triage Staff, and other provider roles), LGU health administrators, and registered NCHO patients requiring outpatient services.

Functional Requirements

- Unified Secure Identity Provisioning: Single login with role handling. Provider access requires credential authentication and provider role authorization. Patient accounts proceed through verification status states before full operational use.
- QR Code Gatekeeping: Patient QR token is generated and stored in QR_ACCESS_TOKENS, returned to the patient passport, and scanned by authenticated providers to resolve patient identity in real time.
- Digital Health Passport & Transparency: Patient dashboard is view-only and displays profile, encounters, records, and privacy logs sourced from live backend data.
- Real-Time Clinical Encounters: Providers start or resume an encounter session, encode vitals and clinical notes, and finalize the record into completed encounter history.
- Medical Records Archiving: Authorized providers upload official diagnostic files through the upload interface, with every upload tied to patient records and audit tracking.
- Automated Audit Logging: Provider-facing access and transaction endpoints are logged by middleware and persisted to AUDIT_LOGS for traceability and patient transparency views.

Non-Functional Requirements

- System Security and Data Privacy: Enforces authenticated sessions, role-based route

protection, provider accountability logging, and backend-managed timestamps for auditable events.

- Performance and Responsiveness: Supports fast QR scan resolution, patient lookup, and encounter workspace transitions during high OPD volume.
- Usability and Accessibility: Separates patient read-only views from provider operational interfaces to reduce input error and speed up consultations.
- System Reliability: Clinical actions persist through backend transactions, with draft support on encounter workspace and resumable in-progress encounters.

Modules

1. Identity and Provisioning Engine

A. Unified Login & Role Selection:

- I. Select role as Patient or Healthcare.
- II. Input email and password credentials.
- III. Authenticate against role-specific account source (PATIENTS or PROVIDERS).

B. Provider Authentication:

- I. Provider logs in through /auth/login with role = provider.
- II. On successful credential match, issue JWT session token with provider identity and role type.

C. Patient Shell Account Creation:

- I. Register using email and password.
- II. Capture required demographic and LGU profile fields during registration.
- III. Auto-create patient_id and issue JWT token for immediate patient session.

D. Triage Verification:

- I. Initial registration data is stored in the patient record.
- II. Provider can update baseline fields during verification/intake workflows.
- III. Verification state is managed through patient verification_status where applicable.

Identity and Provisioning Engine User flow:

This module is the access and account-entry layer of the system. It handles role-based login, patient registration with required demographic inputs, and JWT session issuance used by protected patient/provider routes.

Identity and Provisioning Engine

2. Authorization and Access Module

A. Access Triggering:

- I. Standard trigger (Doctor scans activated live app or printed QR code)
- II. Manual failsafe trigger (Provider lookup by Patient ID or registered email)

B. Identity Verification:

- I. Physical ID verification requirement (for Manual Failsafe)
- II. QR code validity check (for Standard trigger)

C. Token Management & Security:

- I. Data Privacy Act compliance enforcement
- II. QR_ACCESS_TOKENS token lookup and active-state validation
- III. Expires-At timestamp enforcement at backend validation

D. Database Integration

I. QR_ACCESS_TOKENS table transaction logging

Authorization and Access User flow:

This module resolves patient identity for provider workflows. The patient QR token is validated against active and expiry state, then used to open provider clinical record view under authenticated provider session control.

Authorization and Access Module

3. Patient Health Passport Module (View-Only Dashboard)

A. Secure Authentication and Status Verification

- User login and active session validation
- Account status check (Active or Pending)
- Access denial and NCHO verification prompt for pending accounts

B. Primary Health Summary (Static Interface)

- Active QR code display using backend-issued token string
- Personal identity display (First Name, Last Name, and Date of Birth)
- Critical baseline data visibility (Blood Type, Allergies, and Active Condition)

C. Expandable Medical Intelligence (Accordion/Sectioned Interface)

- Medical Records section (Uploaded records and supporting clinical files)
- Visit History section (Chronological list of past encounters)
- Privacy and Access Log (Audit trail of access events)

D. Clinical Accuracy Enforcement

- Strict read-only state for all patient-facing components
- No create, update, or delete capability available in patient routes

E. Backend Data Synthesis

- Consolidated retrieval through backend joins and includes
- Real-time fetching from PATIENTS, ENCOUNTERS, and MEDICAL_RECORDS
- Audit trail retrieval from AUDIT_LOGS

Patient Health Passport User flow:

This module is a read-only patient experience. The QR code shown in the passport maps to the patient token in QR_ACCESS_TOKENS, and provider scan resolves identity without exposing any patient-side write operation. Patients can review what exists in the system and who accessed it, but cannot alter clinical records.

Patient Health Passport User flow

4. Clinical Triage and Encounters Module

A. Session-Based Clinical Workflow

- Provider enters through QR scan or manual lookup, then opens Patient Record View
- Provider explicitly starts or resumes an encounter session before encoding clinical data
- Encounter session is bound to one patient and one authenticated provider context

B. Real-Time Vitals and Assessment

- Structured encoding of Blood Pressure (systolic/diastolic), Temperature, and Weight
- Formal documentation of Chief Complaint as reported during consultation

C. Diagnosis and Treatment Planning

- Clinical Diagnosis entry is required to complete an encounter
- Treatment Plan and e-prescriptions are encoded during encounter finalization

D. Data Integrity and Validation

- Required-field validation for Chief Complaint and Diagnosis before finalize
- Encounter status lifecycle enforced (IN_PROGRESS to COMPLETED)
- Optional first-encounter baseline capture (Blood Type and Allergies) persists to patient profile

E. Database Integration

- Encounter writes persist to ENCOUNTERS
- Prescription entries persist to PRESCRIPTIONS linked to encounter
- Audit logging is triggered for start, access, finalize, and cancel routes

Clinical Triage and Encounters User flow:

Live workflow is: Provider scans QR or performs lookup -> opens patient clinical view -> starts or resumes encounter -> encodes vitals and clinical notes in the encounter workspace -> finalizes to completed encounter. Lab result files are not entered in this module; they are handled through the dedicated upload workflow to preserve clinical record separation and upload control.

Clinical Triage and Encounters User flow

5. Medical Records Archive Module

A. Asynchronous Data Management

- Secure repository for non-encounter files (e.g., X-Rays, CBC results, diagnostics)
- Chronological storage independent from active consultation forms

B. Official Documentation Workflow

- Authorized provider upload interface for medical files
- Result context captured through required metadata fields

C. Data Integrity Protocols

- File format validation for PDF and image uploads
- Upload-centered workflow for official document entry

D. Patient Transparency Access

- View-only patient access to uploaded records
- No patient-side file alteration or deletion capability

E. Database Integration

- Inserts into MEDICAL_RECORDS with patient/provider linkage
- Upload and access actions captured in audit trail events

Medical Records Archive User flow:

This module archives official clinical files through controlled provider upload workflows and displays the resulting records to patients in read-only format.

Medical Records Archive User flow

6. Automated Audit Logging Module

A. Invisible Surveillance Layer

- Middleware-level execution on protected provider endpoints with no manual trigger

requirement

- Automatic capture of provider record access, encounter actions, and upload actions

B. Data Capture and Attribution

- Provider identity captured from authenticated session context
- Patient identity resolved from route params/body/request context and linked to event
- Action narrative recorded through explicit action descriptions and endpoint fallback data

C. Temporal Accountability

- Server-generated event timestamp recorded for every audit transaction
- Audit entries are produced post-response completion to reflect completed actions

D. Transparency Integration

- AUDIT_LOGS powers patient-facing Privacy and Access Log views
- Admin monitoring endpoints consume the same audit source of truth

E. Database Integration

- Automated insertion into AUDIT_LOGS
- Sequential log identifiers maintain continuous accountability chain

Automated Audit Logging User flow:

This module runs continuously in the backend request pipeline. It is mandatory for provider workflow observability, does not rely on user-triggered logging actions, and feeds both patient transparency interfaces and admin-level audit review.

Clinical Triage and Encounters User flow

Business Requirement

(Diagram and artifacts unchanged)

Entity Relationship Diagram

(Diagram unchanged)

User Persona

Sample user persona 1

(Diagram/content unchanged)

Sample user persona 2

(Diagram/content unchanged)

Userflow

Full System Userflow:

Userflow Link

(Diagram unchanged)

Journey Map

Sample Journey Map - User Persona 1

(Diagram unchanged)

Sample Journey Map - User Persona 2
(Diagram unchanged)

Sitemap
Site Map

Style Guide

A style guide is a document that outlines the visual and design elements of an application to ensure consistency in branding and user interface. The visual assets and examples remain unchanged.

Color pallete
(Unchanged)

Button & Badges
(Unchanged)

Typography
(Unchanged)

Cards & Icons
(Unchanged)

Form Element and Spacing System
(Unchanged)

Shadow and Elevation & Border Radius
(Unchanged)

Wireframes

Low-Fi

Landing Page

This serves as the initial entry point for the NCHO Patient-Link web application.

Login

This represents the Unified Secure Identity Provisioning interface with role-aware access (Patient or Healthcare Provider) and credential-based authentication.

Provider Portal

This is the provider operational entry point for patient search, QR scanning, and fast transition to patient clinical records.

Patient Directory

A quick-access list and lookup interface for registered patients, supporting fallback access when QR scanning is not used.

Active QR Session

This screen represents QR scan-based patient identity resolution. An authenticated provider submits the scanned token and opens the corresponding patient clinical view.

Digital ITR (Individual Treatment Record)

This maps to the live encounter workspace pattern where providers start/resume an encounter, encode vitals and consultation notes, and finalize the visit.

Audit Logs

This interface surfaces backend-generated audit events for administrative accountability, including provider identity, patient linkage, action description, and timestamp.

Hi-Fi

Home Page

The branded landing interface for NCHO Patient-Link.

Health Care Provider main page

The provider portal optimized for rapid patient access and consultation flow continuity during OPD operations.

Health Care Provider Patient Record

The clinical record view where providers review baseline data, visits, records, and launch encounter sessions.

Lab Result

Part of the Medical Records Archive, this interface handles authorized provider uploads of official laboratory and diagnostic files.

Diagnoses

A structured encounter component where provider diagnosis and treatment plan are finalized within the active encounter workspace.

Visit History

A chronological list of completed encounters shown in both provider and patient read contexts.

Patient Portal

The patient dashboard acting as a digital Health Passport with read-only access and QR presentation.

Patient Personal Record

A read-only view of baseline and personal medical context, including blood type and allergies when present.

Patient Privacy and logs

A transparency view that displays audit events tied to the patient record.

Development Team and Assigned Module (Unchanged)

Appendixes

USER PERSONA - GRP 42_NCHO Patient- Link_DOCUMENT

USERFLOW - GRP 42_NCHO Patient- Link_DOCUMENT

JOURNEY MAP - GRP 42_NCHO Patient- Link_DOCUMENT

SITEMAP - GRP 42_NCHO Patient- Link_DOCUMENT

STYLEGUIDE - GRP 42_NCHO Patient- Link_DOCUMENT

WIREFRAMES - NCHO PATIENT LINK WIREFRAMES